Project Name: FeedEx (Smart Food Delivery Sorting System)

Team Members:

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Problem (The "Why")

Food delivery orders from multiple employees often arrive at the office at approximately the same time. The packages are usually placed together at a reception or collection area, making it difficult to identify which package belongs to which employee.

Manual identification can be time-consuming and may result in food being picked up by the wrong person, especially when packages have similar appearances or unclear labels.

- **Current Issue:** Multiple food packages are mixed together and must be manually identified by employees or receptionists.
- **Impact:** Employees may receive the wrong food, while receptionists spend additional time identifying and organizing incoming packages.

Proposed Solution

The system provides a simple web-based food delivery management system with camera-assisted package identification.

Employees register their food order in the system before the food arrives. When the package arrives, the receptionist scans the package using a camera. The system attempts to identify the order using a QR code or OCR and then displays the corresponding employee and order information.

The MVP will focus on **identification assistance**, rather than attempting to automatically recognize the type of food or determine ownership based on the package's visual appearance.

- **Our Approach:**
    1. The employee registers an order and receives an Order ID.
    2. The package is scanned using a camera.
    3. The system detects the QR code or reads the Order ID using OCR.
    4. The detected Order ID is matched against the database.
    5. The system displays the employee and order information.
- **Key Value:** Reduce the possibility of food delivery mix-ups while keeping the implementation small enough to be completed within two weeks.

Core Features

- **Feature 1: Order Registration**
  Users can register an incoming food order with basic information such as employee name, Order ID, food description, and delivery platform. The system stores the order and provides its current status.
- **Feature 2: Camera-Based Order Identification**
  The receptionist can use the browser camera to scan an incoming food package. The system attempts to identify the Order ID using QR recognition. OCR can be implemented as a fallback if time permits.
- **Feature 3: Order Verification & Status**
  After an Order ID is detected, the system displays the corresponding employee and order information. The receptionist can verify the result and mark the order as Arrived or Collected.

High-Level Architecture & Quality Assurance

- **Architecture Pattern:** Layered Architecture
- **Frontend:** React
- **Backend:** ASP.NET Core Web API / .NET 8
- **Database:** PostgreSQL
- **Computer Vision:** QR Code detection, with OCR as an optional fallback
- **Design Patterns Used:**

    - Dependency Injection
    - Repository Pattern
    - Service Layer Pattern
    - DTO Pattern
- **Code Quality Goal:** We commit to maintaining a green Jenkins pipeline, writing unit tests to be tracked via Cobertura, and keeping Qodana static analysis warnings below 10 (Target: 0).

Identified Challenges & Risks

- **Risk 1: QR Code or Label Cannot Be Detected**
  The package may not contain a QR code, or the QR code may be covered, damaged, or difficult for the camera to detect. The system should provide a manual Order ID input as a fallback.
- **Risk 2: OCR Accuracy**
  If OCR is implemented, folded plastic bags, poor lighting, small text, and unclear labels may reduce recognition accuracy. Therefore, OCR should be considered a secondary feature rather than the core identification mechanism for the two-week MVP.
- **Risk 3: Real Delivery Platform Integration**
  Direct integration with GrabFood, GoFood, or ShopeeFood may require APIs or permissions that are outside the project's scope. The MVP will use manually registered or simulated order data.
- **Risk 4: Limited Development Time**
  The project is intentionally limited to three core features. Advanced Computer Vision, automatic package sorting, face recognition, delivery-platform integration, notifications, and automatic label printing are excluded from the two-week MVP.